SALES
ANALYSIS
— DASHBOARD —
INSIGHT | PERFORMANCE | GROWTH

Superstore Sales Analytics

End-to-End Data Warehouse & Power BI Documentation

Prepared by: Ahmed Mohamed Ahmed

Date: 31 / 5 / 2026

1. Data Overview

The source data is the SuperStore Orders dataset — a retail sales dataset containing transactional records from a global superstore. The raw data is stored in an Excel file and covers customers, products, orders, shipping, and financials across multiple worldwide markets.

1.1 Dataset Summary

Attribute	Details
Source File	SuperStoreOrders_Clean.xlsx
Total Records	~51,290 rows
Time Period	2011 – 2014
Markets	APAC, EU, US, LATAM, Africa, EMEA, Canada
Categories	Technology, Furniture, Office Supplies

2. Architecture Overview

The project follows a classic multi-layer data warehouse architecture. Data flows from a raw Excel source through four structured layers before reaching the Power BI reporting layer.

Layer	System	Purpose
Source	Excel (.xlsx)	Raw transactional data — no quality control
ODS	SQL Server	Exact copy of source — used for data profiling
STG	SQL Server	Cleaned data + OBT (One Big Table) + individual dimension tables
DWH	SQL Server	Final star schema — Dim tables + Fact table
Power BI	Power BI Desktop	Semantic model, DAX measures, interactive dashboards

The pipeline is orchestrated using SQL Server Integration Services (SSIS) with three packages: ODS.dtsx, STG.dtsx, and DWH.dtsx.

3. Logical & Physical Model

3.1 Conceptual Model

At the conceptual level, the business domain revolves around ORDERS placed by CUSTOMERS for PRODUCTS, which belong to CATEGORIES. Orders are classified by ORDER PRIORITY, associated with a LOCATION (city, state, country, market, region), and analysed over a DATE dimension.

Core business entities:

- Customer — who placed the order
- Product — what was ordered (linked to Category)
- Location — where the order was delivered
- Date — when the order was placed
- Market — the global market / region grouping
- Order Priority — urgency level of the order
- Fact (Orders) — the measurable transaction connecting all dimensions

3.2 Logical Model (Star Schema)

The logical model implements a Star Schema design. A central Fact table holds all measurable KPIs and foreign keys that point to surrounding dimension tables, allowing efficient BI queries without complex multi-hop joins.

Table	Type & Key Attributes
Dim_Customer	customer_id (PK), customer_name, segment
Dim_Location	location_id (PK), city, state, country, region, market
Dim_Category	category_id (PK), category, sub_category
Dim_Product	product_id (PK), product_name, category_id (FK → Dim_Category)
Dim_Market	market_id (PK), market name, region
Dim_OrderPriority	priority_id (PK), order_priority, priority_value, ship_mode
Dim_Date	date_key (PK), Date, Day, Day_Name, Day_Of_Week, Month, Month_Name, Quarter, is_weekend
Fact_Orders	FK to all dimensions + sales, quantity, discount, profit, order_id, ship_date

3.3 Physical Model

The physical model is implemented in SQL Server. Below is the actual model diagram from Power BI showing all tables and their relationships in the final star schema.

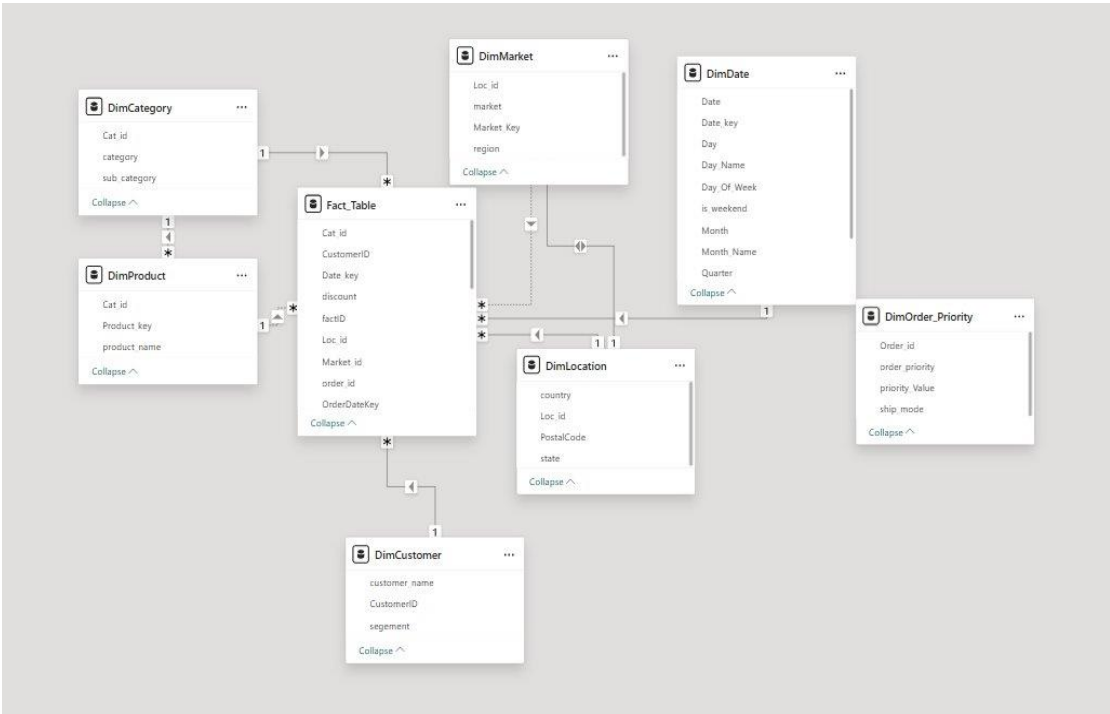

Figure — Power BI Physical Model (Star Schema)

4. ETL Pipeline — Step by Step

The ETL process is broken into three SSIS packages executed in order. Each package is responsible for one layer of the architecture.

Step 1 — Load Raw Data into ODS (ODS.dtsx)

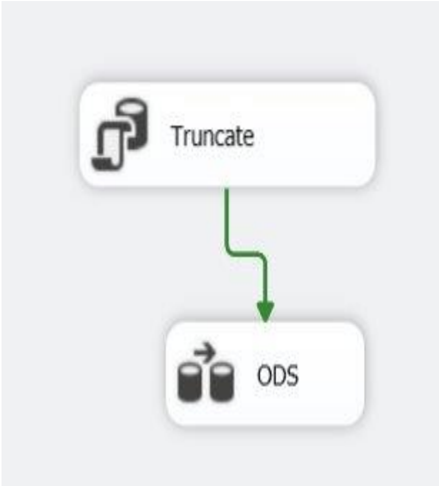

Figure — ODS Control Flow: Truncate → Load ODS

The ODS package wipes the ODS table clean and reloads it directly from the Excel source. It is our raw snapshot used for profiling — detecting nulls, duplicates, wrong types, and bad date formats.

Inside the Data Flow Task, order_date and ship_date are converted from text format (M/D/YYYY) to a proper DATE data type using a Derived Column transformation with SSIS expressions.

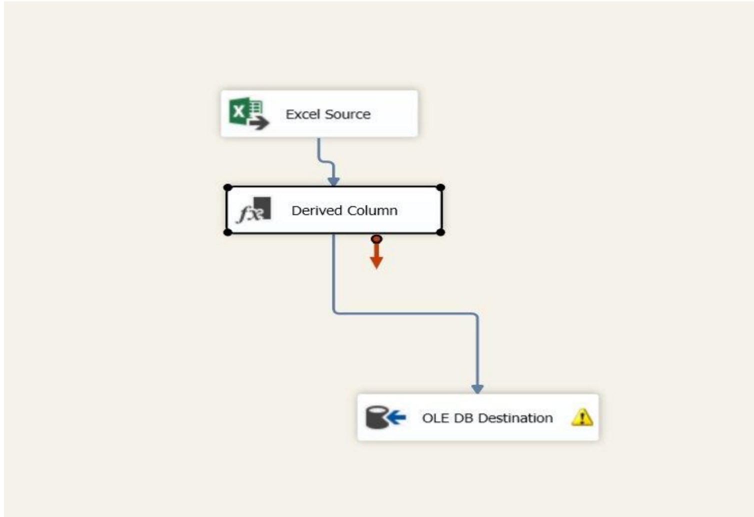

Figure — ODS Data Flow: Excel Source → Derived Column → OLE DB Destination

Step 2 — Build Staging Layer (STG.dtsx)

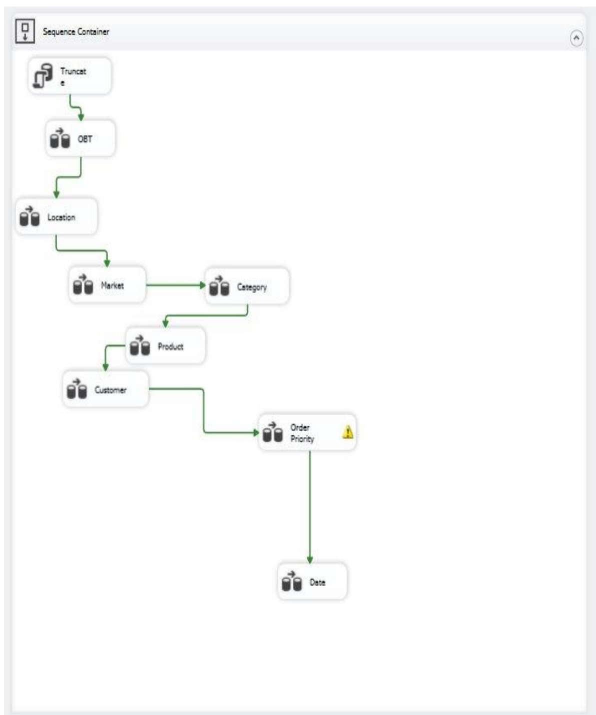

Figure — STG Control Flow: OBT + Dimension Tables

The STG package reads cleaned data from ODS and produces: the OBT (One Big Table) — a fully cleaned denormalized table — and individual dimension tables extracted from it via SELECT DISTINCT queries.

Step 3 — Load Data Warehouse (DWH.dtsx)

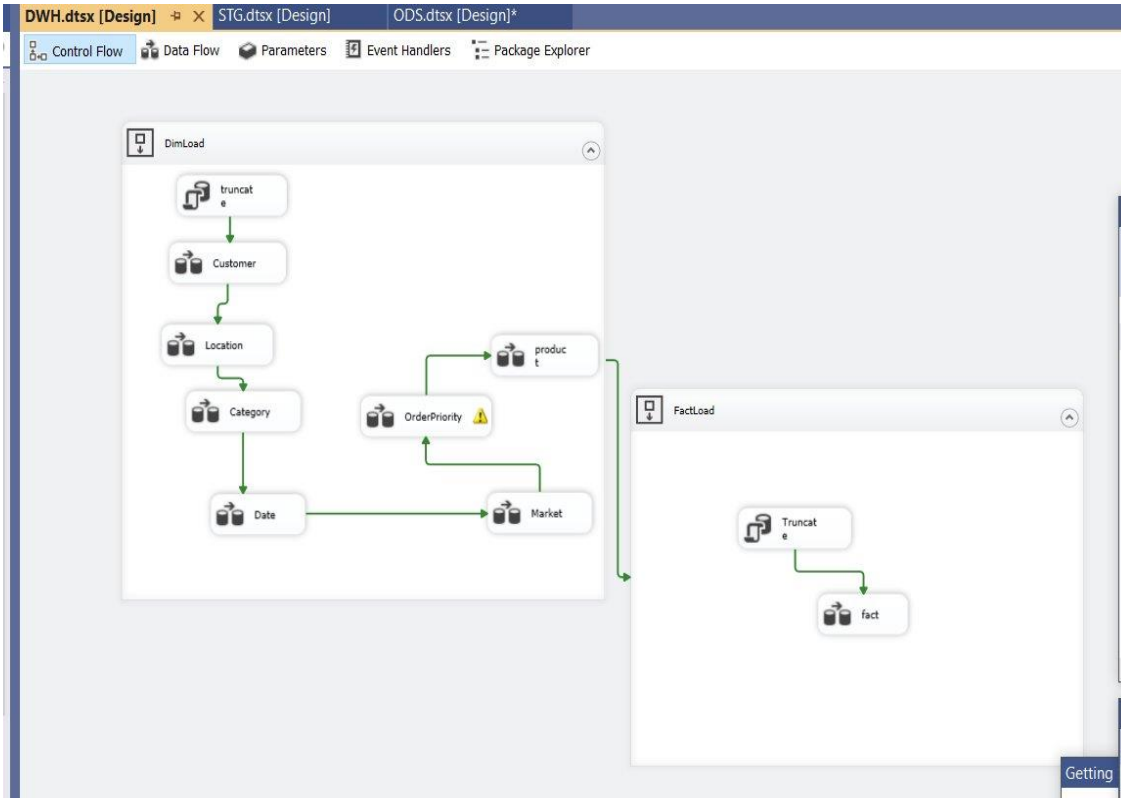

Figure — DWH Control Flow: DimLoad + FactLoad

The DWH package reads from STG and populates the final star schema. It has two Sequence Containers:

DimLoad — Load All Dimensions

Truncates DWH dimension tables and reloads them from their STG equivalents. The sequence respects FK dependencies — Category loads before Product, etc.

FactLoad — Load the Fact Table

After all dimensions are loaded, FactLoad truncates Fact_Orders and reloads it by joining OBT with all dimension tables to resolve surrogate keys.

5. Semantic Model

After the DWH is fully loaded, Power BI Desktop connects to SQL Server and imports the star schema. The semantic model defines relationships and DAX measures on top of the physical tables.

5.1 Model Relationships

All dimension tables relate to Fact_Orders on their surrogate keys with Many-to-One cardinality and single-direction cross-filtering (standard for star schemas).

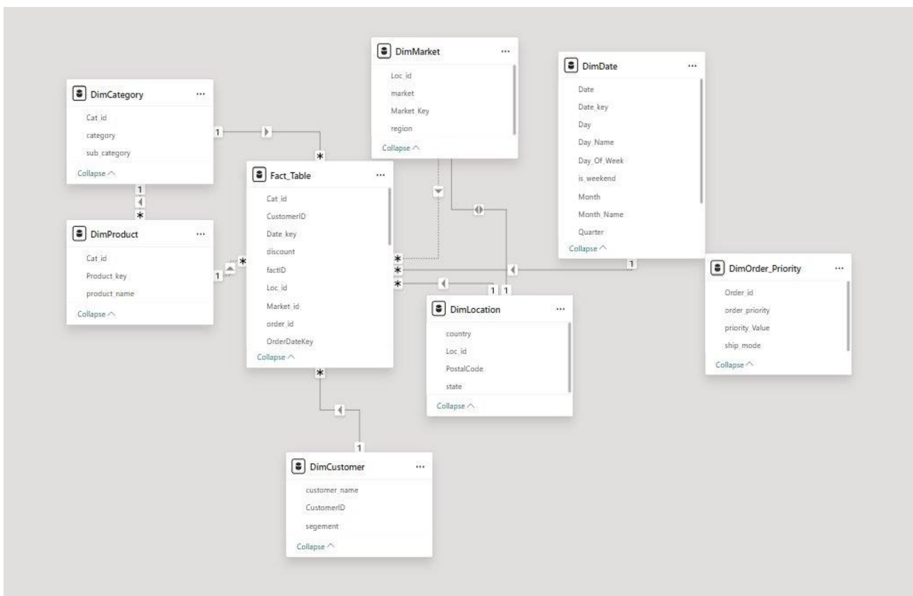

Figure — Power BI Model Relationships

5.2 DAX Measures

The following DAX measures drive all KPI cards and charts in the dashboard:

Measure	DAX Formula	Description
Total Sales	SUM(Fact_Orders[sales])	Total revenue across all orders
Total Profit	SUM(Fact_Orders[profit])	Net profit across all orders
Total Quantity	SUM(Fact_Orders[quantity])	Total units sold
Average Discount	AVERAGE(Fact_Orders[discount])	Average discount rate applied
Profit Margin	DIVIDE([Total Profit], [Total Sales], 0)	Ratio of profit to sales
Customer Sales	SUM(Fact_Orders[sales])	Sales per customer — used in customer bar chart

6. Power BI Dashboard Overview

The Power BI report has two pages: the Executive Dashboard for high-level KPIs and the Profitability & Customer Insights page for detailed analysis.

6.1 Executive Dashboard

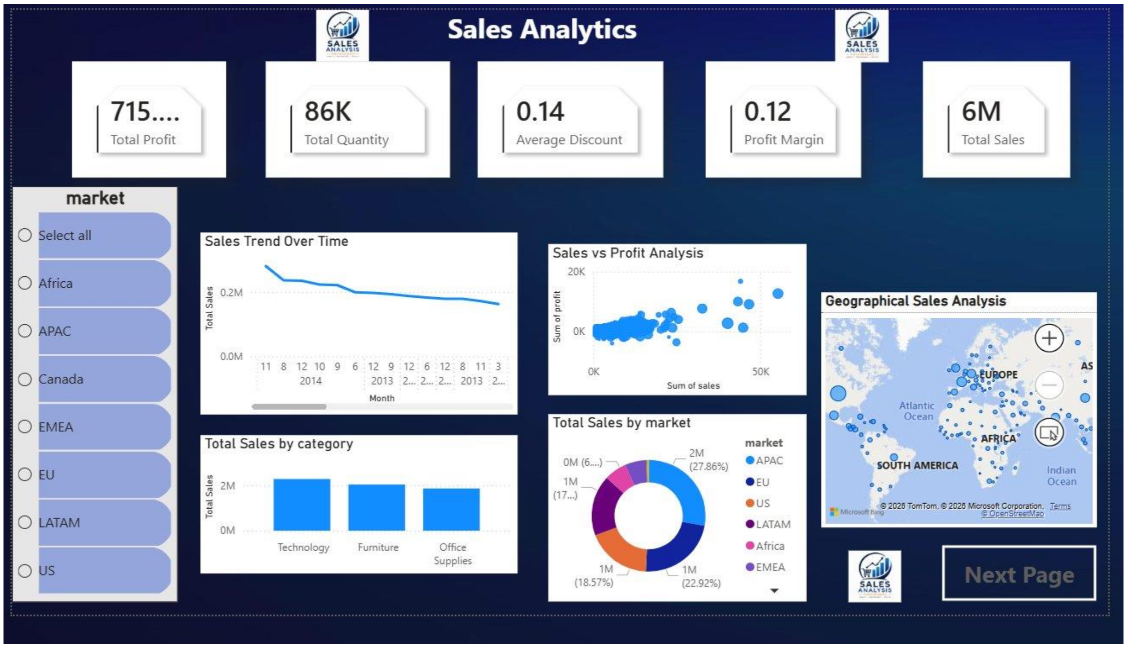

Figure — Executive Dashboard (Page 1)

Visual	Description
KPI Cards (×5)	Total Profit, Total Quantity, Average Discount, Profit Margin, Total Sales
Market Slicer	Filter panel for all 7 markets — drives all other visuals on the page
Sales Trend Over Time	Line chart — X: Month/Year, Y: Total Sales — shows revenue trend 2011–2014
Total Sales by Category	Bar chart — Technology vs Furniture vs Office Supplies
Sales vs Profit	Scatter chart — shows correlation between sales volume and profitability
Total Sales by Market	Donut chart — market share by region
Geographical Map	Bing Maps — bubble size = Total Sales — geographic distribution of revenue

6.2 Profitability & Customer Insights

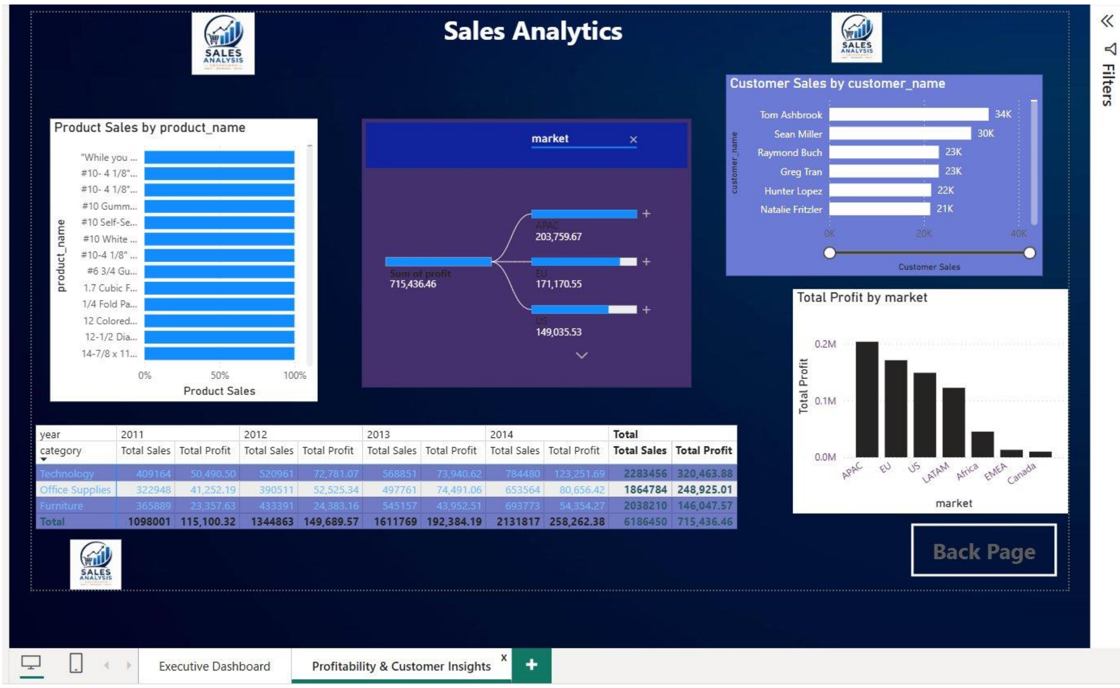

Figure — Profitability & Customer Insights (Page 2)

Visual	Description
Product Sales Bar	Horizontal bar — relative sales by product_name — identifies top-selling products
Market Decomposition Tree	AI visual — breaks Total Profit down by market (APAC: 203K, EU: 171K, US: 149K…)
Customer Sales Bar	Top customers by revenue — Tom Ashbrook 34K, Sean Miller 30K, Raymond Buch 23K…
Total Profit by Market	Vertical bar — profit per market in absolute values — APAC and EU lead
Matrix Table	Cross-tab: Category × Year — Total Sales + Total Profit — year-over-year category performance

---

End of Documentation — Superstore Sales Analytics Project

7. Glossary

Term	Definition
ODS	Operational Data Store — raw landing zone mirroring the source; used for profiling
STG	Staging layer — cleaned data including OBT and individual dimension tables
OBT	One Big Table — denormalized staging table; dimensions are derived from it via SELECT DISTINCT
DWH	Data Warehouse — final star schema with Dim_* tables and Fact_Orders, optimised for analytics
Star Schema	One central Fact table surrounded by Dimension tables — efficient for BI queries
SSIS	SQL Server Integration Services — Microsoft ETL tool used to build and run the pipeline packages
Surrogate Key	Auto-generated integer PK (IDENTITY) assigned in the DWH to each dimension record
Derived Column	SSIS transformation that computes new column values using expressions — used for date conversion
DAX	Data Analysis Expressions — Power BI formula language for calculated measures and KPIs
Fact Table	Central star schema table storing measurable events with FK references to all dimensions
Dimension Table	Lookup table describing a business entity — provides context to Fact data

Thank You.